AROMA PACIFICO

ARCHITECTURE DOCUMENT

Architecture of the pilot

Coffee traceability pilot for Cauca, Huila and Tolima

November 2026

This document presents the architecture proposed to the Federation for the update of the coffee traceability pilot in Cauca, Huila and Tolima. It explains what the pilot needs, the shape of the solution and why it fits a pilot with little connectivity and little budget, and then develops the seven points of the challenge, governance, security, network and connectivity, automation and change, workloads, observability and the design system. It closes with what happens when something fails, how the pilot scales, what it costs, what changes from the 2020 design, the assumptions that shape it and an index of the diagrams and documents that carry the detail.

In the diagrams, blue boxes are what the infrastructure code implements. White boxes are designed and documented, and they stay out of the code.

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1. Terms

Term	Meaning in this document
Node	Autonomous device on a farm that reads probes and publishes readings
Store-and-forward	The device keeps a reading while it has no signal and sends it when the path returns
Data mule	A person or vehicle already traveling a route who carries data from a site with no signal
Case	The record that follows a pest from the alert to the verified fumigation, also called expediente
Trust boundary	The line where the platform stops trusting the network and starts checking identity
Presigned URL	A temporary address that lets one client upload one object, with no stored credential
Landing zone	The set of accounts and rules that exist before any workload
Brigade	One pilot and one support person with a spraying drone and a mapping drone, one per department
Covered by design	Specified in a document with its safe behavior, with nothing deployed for it yet

2. What the pilot needs

The pilot follows coffee from the farm to the buyer in three departments. Five capabilities carry over from the design of 2020, and the connectivity assumption behind that design no longer holds.

Capability	What it means in numbers
Crop and drying monitoring, with an alert to the farmer	200 farms, with a reading every 4 hours in cultivation and every hour in drying. The alert arrives in under 5 minutes with a delivery path, and within 24 hours with no signal
Harvest and packing record	Up to 400 sacks a day at each of the 3 centers
Trip tracking	60 carriers, one position every 5 minutes
Sack lookup	Peaks of 30 reads a minute in harvest, available within 4 hours of a failure
Drone service	Five services that do not replace one another

Two requirements apply across all of them.

Requirement	What it means in numbers
Retention	Five years for readings, photos, flights, sack history and personal data
Recovery	Sack events lose at most 15 minutes

The hard part is the field. Farms have no cable, some have no signal, drones fly far from any phone, and each center starts with no network at all. The load itself is small, and the design treats it that way.

3. The solution in one page

Sites at the edge hold their data until a path exists. A serverless platform in AWS receives each event, stores it once, and runs a function that ends when its result is written. Between events, nothing runs and almost nothing is billed.

Zone	What lives there	Key decision
Field	Solar nodes, LoRa nodes, shared gateways, drones and phones	Each device buffers locally and needs no local server
Collection centers	One LAN pattern repeated three times, with perimeter firewall and a dedicated 100 Mbps WAN	The second octet is the only difference between centers
Governance	Seven accounts under one AWS Organization	Controls apply from the root, and every account inherits them
Production	IoT Core, API Gateway, functions, queues, DynamoDB, S3, Bedrock	No servers, no instances, no idle compute
Network	One VPC per environment, with seven subnets and a VPN per center	The VPC exists for the path from the centers
Security and observability	KMS, Secrets Manager, CloudTrail, GuardDuty, CloudWatch	Separate account, so a compromised workload cannot erase its trail
Delivery	Repositories, pipeline, remote state	Plan, a person approves, then apply
Second cloud	Azure East US	Prepared with contracts and nothing running

Why this shape

Event-driven and serverless. The load is about 11,000 messages a day in harvest, counting device readings and carrier positions, which is 0.13 messages a second. A server sized for that would sit idle almost all the time, and the budget of the pilot cannot justify it. Persistent compute waiting for objects is also ruled out for the pilot.

Buffer at the edge. Coverage cannot be bought with the budget of a farm. A device that holds its data for up to 24 hours turns a connectivity problem into a delay, and the alert path tolerates that delay by definition.

One regional endpoint. Every farm publishes to the same address. Adding a farm means adding a certificate, with no change to the network, which is what lets the pilot grow to 1,000 farms with no redesign.

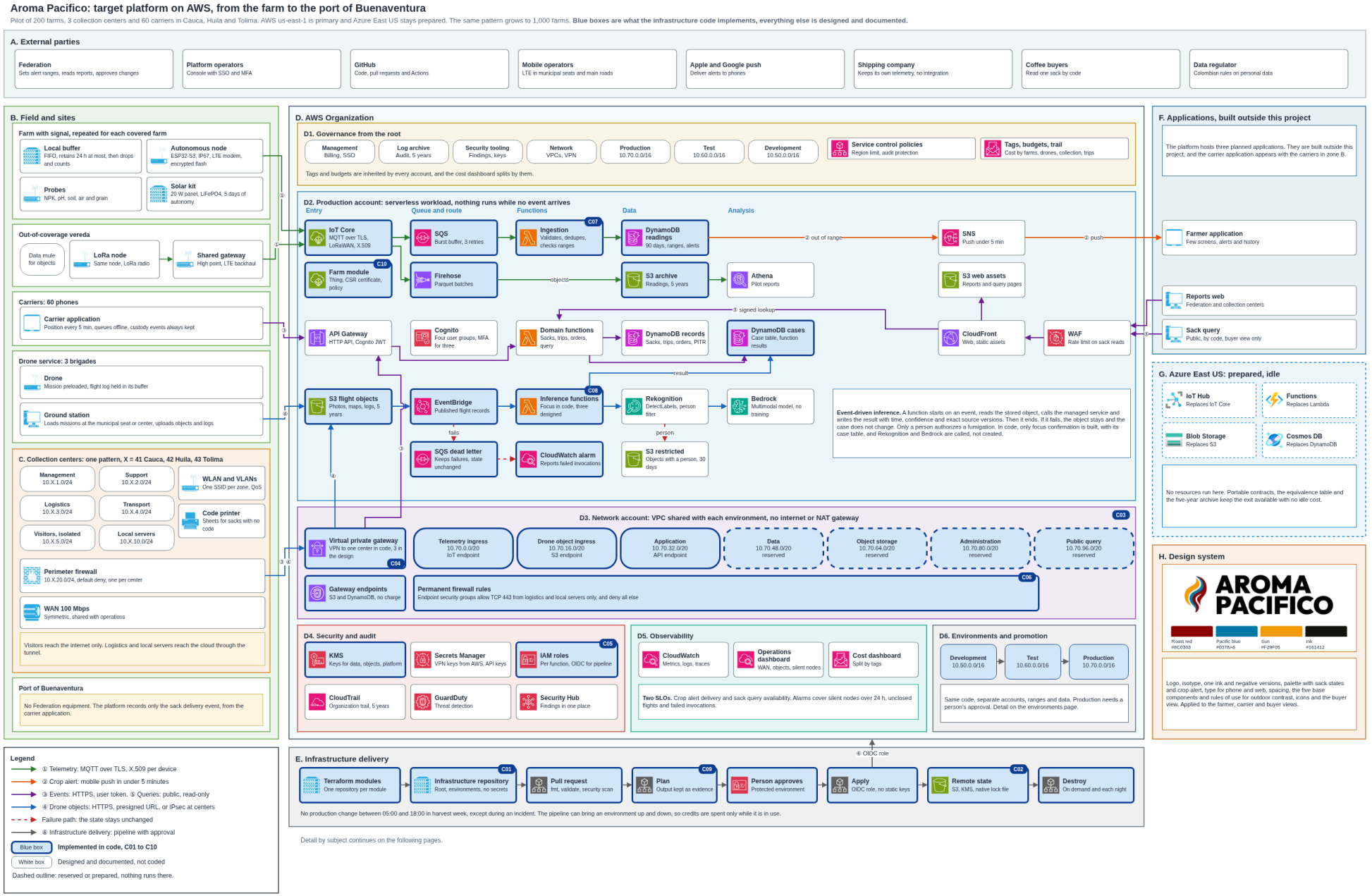
Managed inference. No model is trained. A function starts on a verified flight record or a completed telemetry object, reads the exact stored version, calls a managed service and writes an idempotent result. Replacing the model later touches one configuration value.

A person decides. Functions confirm, classify and estimate. Only a named person authorizes a fumigation, and no trigger can do it. This follows from the risk of the action, since a wrong fumigation cannot be undone.

Key decisions

Each choice has a decision record with its alternatives, and the decision index lists all of them. These are the ones that shape the platform most.

Subject	Decision	Why
Primary and second cloud	AWS `us-east-1`, with Azure East US prepared and idle	Managed IoT, serverless and inference services fit a small, intermittent load, and the second cloud costs nothing while idle
Out-of-coverage farms	A LoRaWAN gateway shared by the vereda for readings, and a data mule for objects	No new base station and no weekly visit
Mobile network generation	No dependence on 2G or 3G, and the modem chosen with the operator	The access network keeps changing during the pilot
Farm node connection	One cycle each hour, configuration through a Device Shadow	A sleeping node saves energy and still receives its configuration
Telemetry storage	DynamoDB for 90 days and S3 for five years	Fast reads for recent data and cheap retention for the rest
Sack event history	The sack state is derived from its insert-only events	A late or repeated event can never corrupt a sack
Drone operating mode	Area work for polygons and estimates, precision work for foci	The drone flies where a sensor points, which costs far less than sweeping farms
Flight log and fumigation	One stored log before authorization and another before closure	No fumigation starts or closes on a missing record
Managed inference	Bedrock for analysis and Rekognition as a person filter	No model is trained, and no person reaches the model
Inference failure policy	Four receives, a dead letter queue for each function and rule, no state change on failure	A failure leaves the case as it was and is visible to an operator
Center cloud link	Static site-to-site VPN on a virtual private gateway, with a dedicated link and shared fiber as fallback	Center traffic stays private at a fixed monthly cost
Public lookup path	Function URL behind CloudFront with a web firewall, capped by five reserved copies	The only public service has a ceiling on load and on cost
Infrastructure code scope	Telemetry ingestion and the focus confirmation function as the coded patterns	They prove the critical paths within the 100 hours



The complete platform, from the farm to the port of Buenaventura. Section 17 shows where each required element appears.

4. The six flows

Flow	Path	Identity	Protocol
1. Telemetry	Node, IoT Core, SQS, ingestion function, DynamoDB, with a second copy through Firehose to S3	X.509 certificate per device	MQTT over TLS
2. Crop alert	Ingestion function, SNS, farmer's phone	IAM role of the function	Mobile push
3. Sack, trip and order events	Application, API Gateway, domain function, DynamoDB transaction	Cognito token, MFA at centers and for drones	HTTPS, or IPsec from a center
4. Drone objects	Ground station, S3, EventBridge, work queue, inference function, Rekognition, Bedrock, case	Cognito token and presigned URL	HTTPS, or IPsec from a center
5. Public query	Browser, CloudFront with WAF, lookup function through its function URL	None, read by code	HTTPS
6. Infrastructure delivery	Pull request, plan, approval, apply	GitHub OIDC role	HTTPS

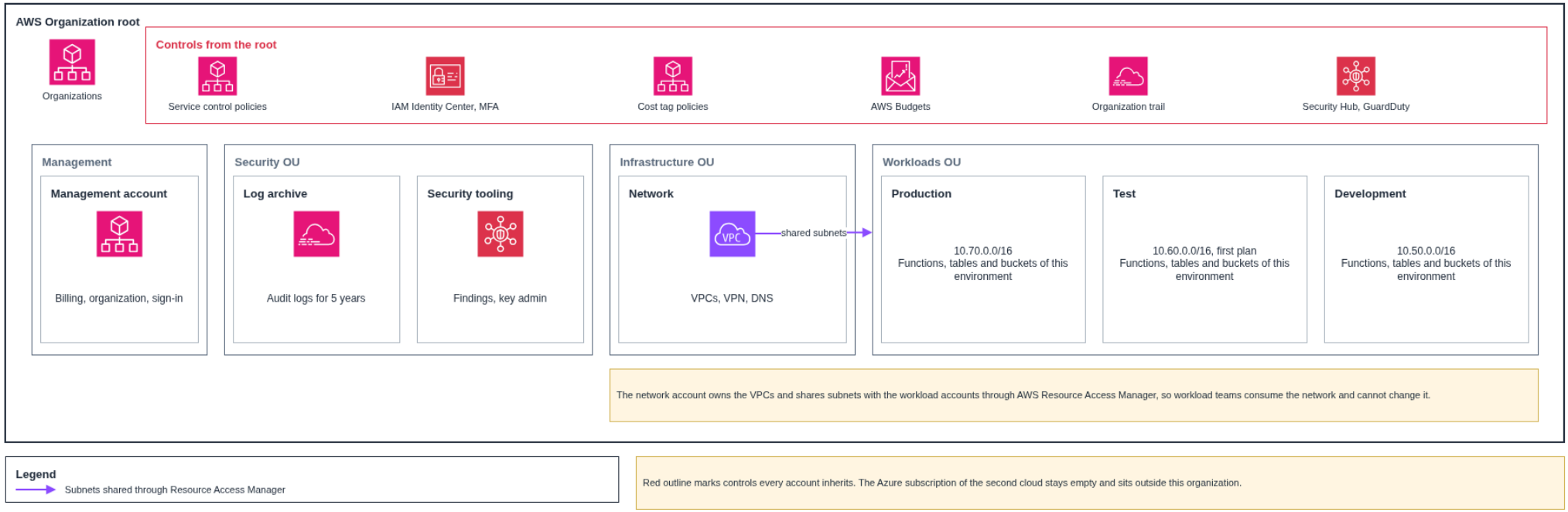
The buyer flow deserves a note. A buyer does not sign in, so the sack code works as a capability to read one sack. The code is random and reveals nothing, and the response holds only the buyer view. Every read is audited.

5. Governance

Every account is born with its rules. The landing zone creates seven accounts under one AWS Organization, and the controls apply from the root, so no team has to remember them.

Account	Contents
Management	Organization, consolidated billing and IAM Identity Center
Log archive	Organization trail and audit logs, five years with object lock
Security tooling	Security Hub, GuardDuty and key administration
Network	VPCs, VPN, DNS and endpoints, shared with each environment
Production, Test and Development	One workload account each, with its own address range

AWS Organization: accounts, boundaries and controls applied from the root



Accounts of the organization and the controls that apply from the root.

Control	What it stops
Region limit	Regional resources outside `us-east-1`
Audit protection	Stopping or deleting the organization trail
Encryption	Volumes or buckets without default encryption
Root user and long-lived users	Sign-in without MFA and without a named person
Tags and tag policy	Resources without `Client`, `Project`, `Environment`, `Owner`, `CostCenter` and `ManagedBy`
Budgets	Silent overspending, with alerts at 80 and 100 percent

The `CostCenter` tag carries farms, drones, collection, trips and tests, so the cost dashboard splits the month by those groups. The production budget is USD 230 a month and the test and development budgets USD 15 and 10.

The audit of sensitive events records who read a sack, who changed an alert threshold, which device published telemetry and who delivered a flight object, and keeps the evidence for five years in the log archive account, where no workload account can alter it. The requirement matrix of section 15 records what stays from the 2020 design and what leaves. AWS `us-east-1` is the primary cloud and Azure East US stays prepared with nothing running, as the cloud selection explains.

The demonstration runs on one account with the environments separated by name, address range, state and pipeline role, and the pilot uses the seven accounts.

6. Security

Each actor has its own identity with the least access its work needs, and people, services and devices never share credentials. The identity and access design defines six human roles.

Role	Identity	Strong authentication
Farmer	Cognito, farmer group	Phone number, with a session of 30 days
Carrier	Cognito, carrier group	Phone number, with a session of 30 days
Center operator	Cognito, center group	MFA
Buyer	None, the sack code is the capability to read one sack	Does not apply
Drone operator and Federation drone supervisor	Cognito, two separate groups, and no account holds both	MFA
Platform operator	IAM Identity Center with temporary credentials	MFA

Services use IAM roles with one role for each function, and each farm node and drone carries its own X.509 certificate. A node or a drone that is retired or lost has its certificate revoked and stops publishing, as the encryption, secrets and revocation design defines. Every flow travels over TLS, readings and objects rest encrypted with KMS keys, and secrets live in Secrets Manager, outside repositories and outside the infrastructure state.

The sack code is random and reveals nothing about the farmer. The buyer reads origin, process dates, state and drying, the center reads the trip and the farmer reads the sacks of their own farms, as the views by actor define. Personal data of farmers and carriers stays five years and is deleted after that, under the personal data and retention rules. Crop images leave Colombia only for the managed inference in the United States, and an image that shows a person never reaches the model, as the inference Region and data leaving the country states.

The firewall policy is one matrix of every rule at the centers, the tunnel, the VPC and the public edge, each with an identifier and whether it stays for the pilot or is temporary.

7. Network and connectivity

Each kind of site has its protocol, its local buffer and its behavior with no signal.

Flow	Protocol out	Local buffer	With no signal
Farm with signal	MQTT over TLS on LTE	24 hours at most	Stores, retries each hour, and the alert goes out when the buffer drains
Farm out of coverage	LoRa to a shared gateway, TLS on LTE from there	24 hours at most	Objects travel by data mule on daily routes
Drone	HTTPS from the ground station	Drone buffer	Mission stays loaded, and the log publishes when the link returns
Carrier	HTTPS on LTE	Application queue	Positions queue for 24 hours at most, and custody events stay until the platform confirms them
Collection center	IPsec VPN over a 100 Mbps link	Application queue	Reception continues locally and synchronizes on return
Port	HTTPS on LTE, from the carrier application	Application queue	Only the delivery event is recorded, and it waits in the queue with no time limit

The farm connectivity, out-of-coverage LoRa path, drone mission link, carrier tracking and port delivery designs detail each row. No part of the design depends on a mobile generation that an operator is retiring, as the mobile network validity shows.

Store-and-forward on the farm node. The node writes every reading to encrypted flash before its modem powers on, so a failed connection loses nothing, as the farm connectivity design defines.

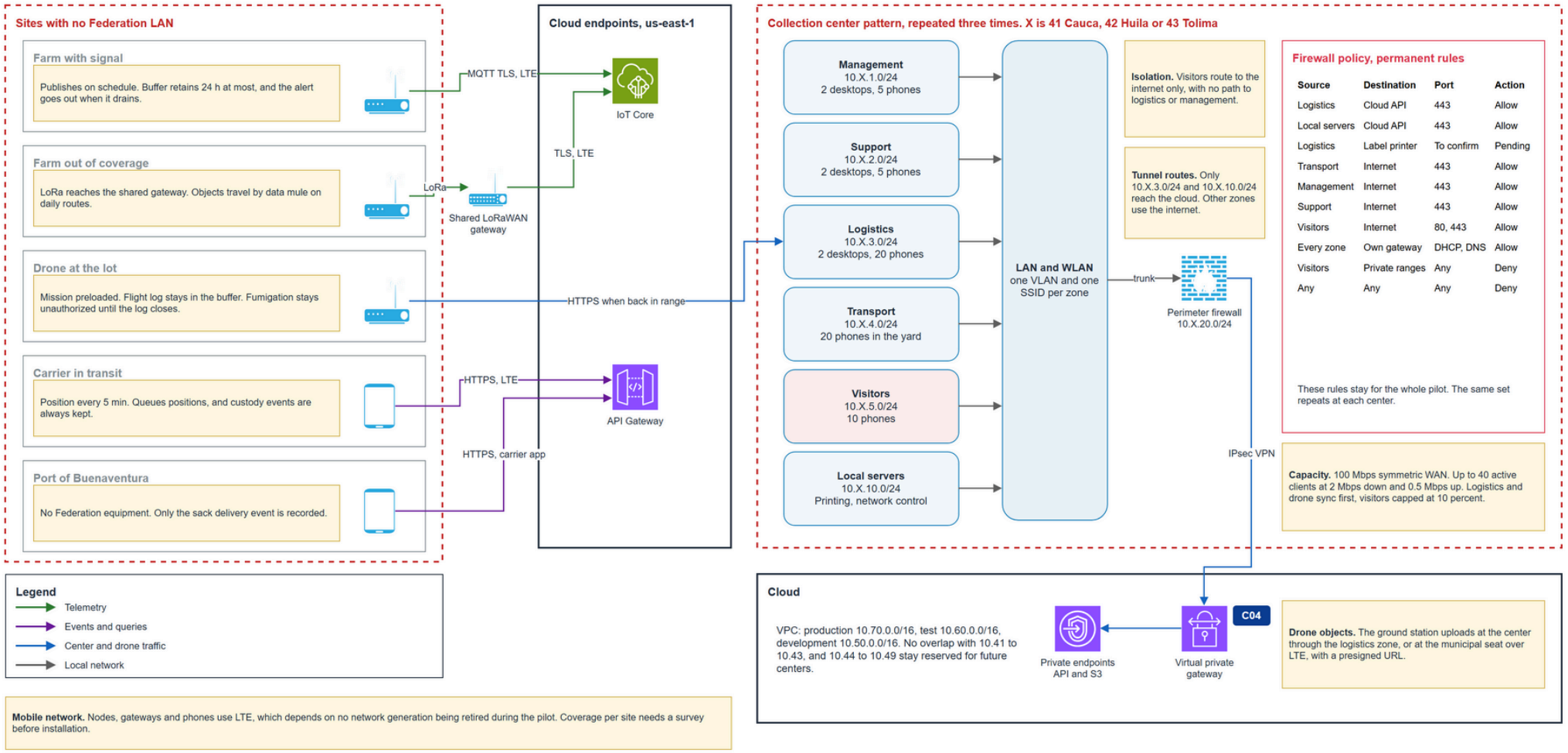
Rule	Behavior
Write first	The reading reaches the flash before the modem powers on
Purge before connecting	Each hourly cycle deletes records of 24 hours or more, then connects
Count what is deleted	Each deleted record adds one to `dropped_total`, so the platform knows what is missing

Rule	Behavior
Drain oldest first	Up to 10 unconfirmed messages at a time
Delete on confirmation	A record leaves the buffer only after IoT Core confirms it
Sequence survives restarts	Duplicates and gaps stay detectable after a reboot

Length of the outage	What reaches the platform	Delay
Under 1 hour	Everything	1 hour at most
1 to 23 hours	Everything	The outage, plus up to 1 hour
Over 24 hours	The readings of the last 23 hours, and the older ones counted	Up to 1 hour after the signal returns

The platform evaluates each reading on arrival, so an alert from a buffered reading goes out at the drain, within the 24-hour limit, and it carries the time of the reading.

How each site connects: protocol, buffer and behavior with no signal

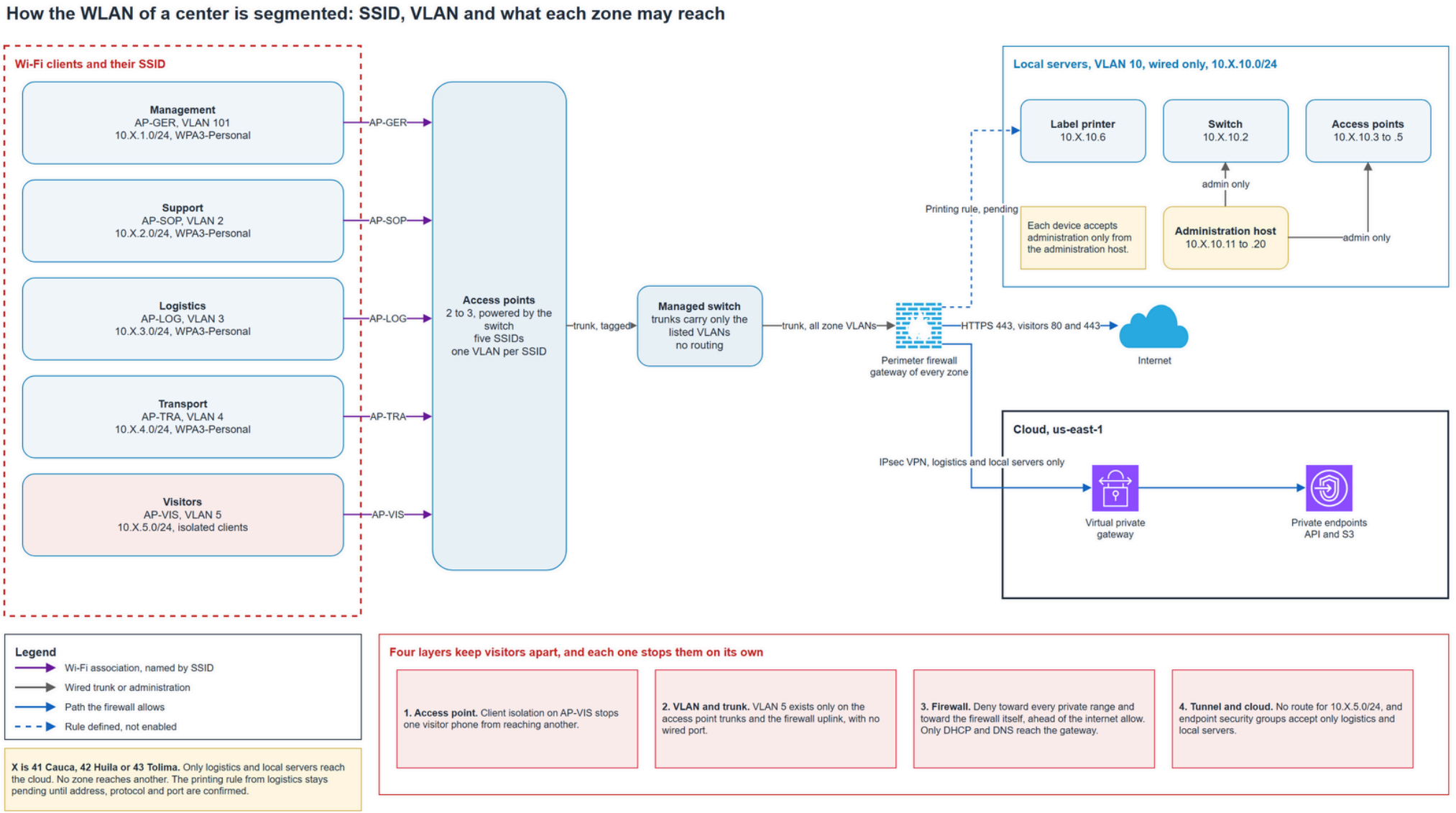


How each site connects and what happens with no signal.

Collection centers. Each center starts with no network, and one LAN pattern repeats in Cauca, Huila and Tolima, changing only the second octet, 41, 42 or 43.

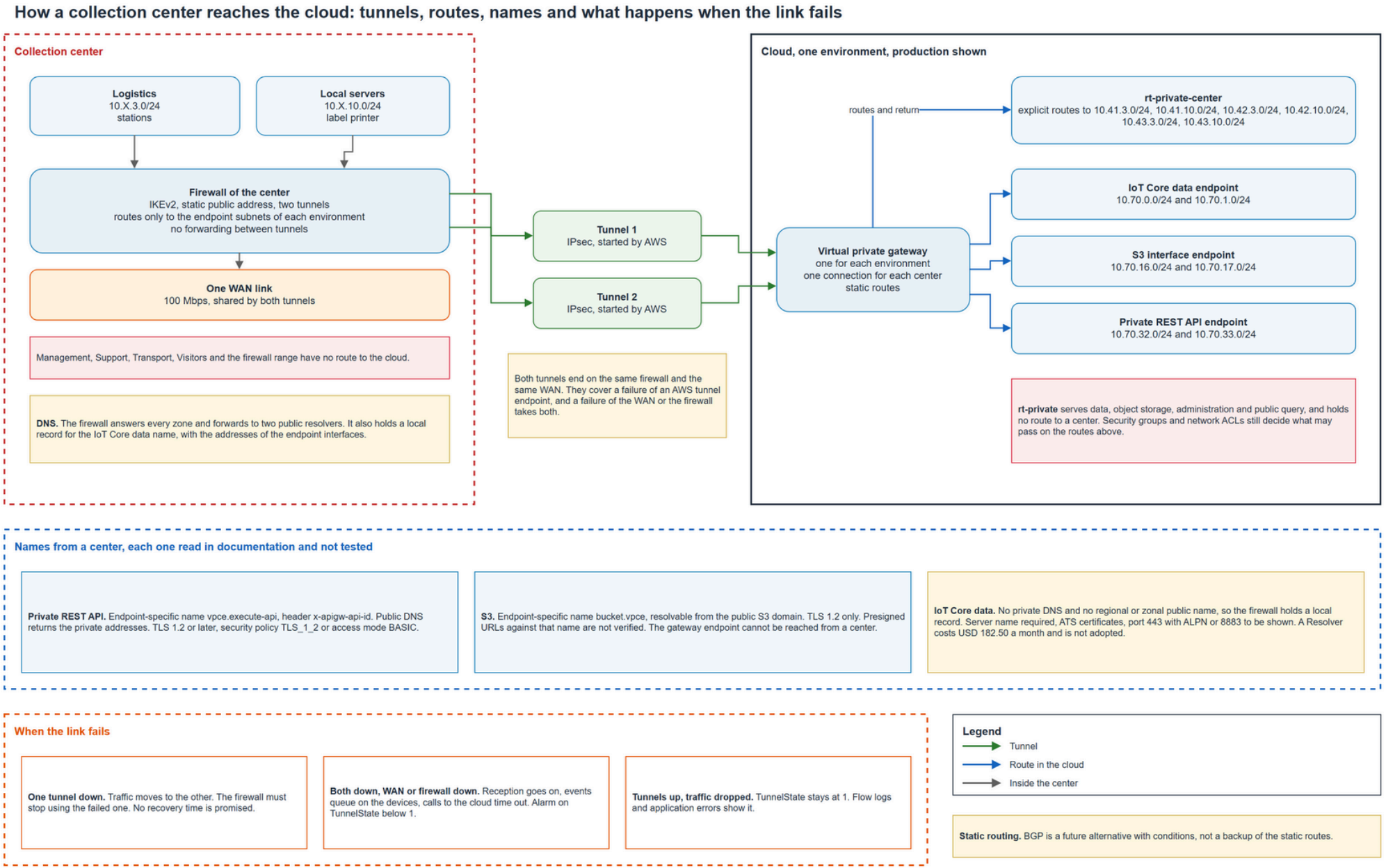
Zone	Network	Reaches
Management	10.X.1.0/24	Internet over HTTPS
Support	10.X.2.0/24	Internet over HTTPS
Logistics	10.X.3.0/24	The cloud API through the tunnel
Transport	10.X.4.0/24	Internet over HTTPS
Visitors	10.X.5.0/24	Internet only, isolated and capped, with no route to logistics or management
Local servers	10.X.10.0/24	Printer and equipment management, and the cloud API through the tunnel
Link to the cloud	10.X.20.0/24	Perimeter firewall, default deny

Each zone has its own SSID on its own VLAN, as the center WLAN defines, and each active client gets 2 Mbps down and 0.5 Mbps up, with traffic classes that cap what is not operational. The link of each center is a dedicated 100 Mbps symmetric WAN, with shared business fiber as the fallback, sized in the center WAN sizing.



Which SSID lands on which VLAN, and what each zone can reach.

Link to the cloud. Each center reaches the cloud through a static site-to-site VPN with two tunnels on the virtual private gateway. Only logistics and the local servers reach the private endpoints of the platform, and the second tunnel takes over when one fails, as the center cloud link defines.



Routes, names and tunnels between a center and the cloud, and what happens when the link fails.

Firewall policy. Traffic between a center and the cloud meets independent barriers, the perimeter firewall of the center, the tunnel routes, the security groups of the endpoints and the network ACLs of the subnets, and the public lookup has its own web firewall. The firewall policy lists 38 rules. This summary groups them by enforcement point.

Rules	Source	Destination	Port	Action	Duration
CTR-01, CTR-02	Logistics `10.X.3.0/24` and local servers `10.X.10.0/24`	Endpoint subnets of the environment	TCP 443	Allow	Permanent
CTR-03	Logistics	Label printer in `10.X.10.0/24`	One port, set by the printer model	Allow	Pending
CTR-04 to CTR-06	Management, support and transport	Internet	TCP 443	Allow	Permanent
CTR-07	Visitors `10.X.5.0/24`	Internet	TCP 80 and 443	Allow	Permanent
CTR-09, CTR-10	Visitors	Private ranges and the firewall itself	Any	Deny	Permanent
CTR-11 to CTR-13	Public address of the firewall	Outside addresses of both tunnels	UDP 500, UDP 4500, ESP	Allow	Permanent
CTR-16	Administration hosts `10.X.10.11` to `10.X.10.20`	Firewall administration	TCP 443 and 22	Allow	Permanent
CTR-17	One administration host	Update service of the firewall vendor	TCP 443	Allow	Temporary, one maintenance window
CTR-18	Any	Any	Any	Deny and log	Permanent
TUN-01 to TUN-04	Logistics and local servers only	Endpoint subnets only, through the virtual private gateway	Route	Allow, and no route for any other range	Permanent
SG-01 to SG-03	Center source ranges	IoT Core, S3 and private API endpoints	TCP 443	Allow	Permanent in production, temporary in development and test
SG-04, SG-05	Endpoints and the default group	Any	Any	Deny	Permanent
NACL-01 to NACL-04	Center source ranges	Endpoint subnets, with the reserved subnets closed	TCP 443 in, ephemeral ports back	Allow, and deny everything else	Permanent

Rules	Source	Destination	Port	Action	Duration
WAF-01	Any address	Sack lookup	More than 300 requests in 5 minutes	Block	Permanent
WAF-02, WAF-03	Any address	Public edge	Common attacks and known bad inputs	Block	Permanent

The infrastructure code builds the cloud rules, SG-01 to SG-05, NACL-01 to NACL-04 and the cloud routes of TUN-03. The center rules load from one installation template that is the same for the three centers. Farm nodes, drones away from a center and carriers reach public endpoints, where device certificates, Cognito tokens and TLS are the control. A temporary rule needs an owner, a reason and an end date, and no rule changes in production from 05:00 to 18:00 in a harvest week.

Cloud addressing. Each environment has its own /16 outside the centers' ranges, and every one holds the seven required subnets. The cloud addressing plan grows from 200 to 1,000 farms with no change, since farms hold no VPC addresses.

Subnet	Production	Purpose
Telemetry ingress	10.70.0.0/20	Interface endpoint for IoT Core data
Drone object ingress	10.70.16.0/20	Interface endpoint for S3, for uploads from a center
Application	10.70.32.0/20	Interface endpoint for the private API of the center stations
Data	10.70.48.0/20	Reserved
Object storage	10.70.64.0/20	Reserved
Administration	10.70.80.0/20	Reserved
Public query	10.70.96.0/20	Reserved

Development uses 10.50.0.0/16 and test 10.60.0.0/16 with the same layout. The three data paths are distinct. Farm telemetry enters through IoT Core, carrier positions through the HTTP API, and drone objects through S3 with presigned URLs, and each path is in section 4.

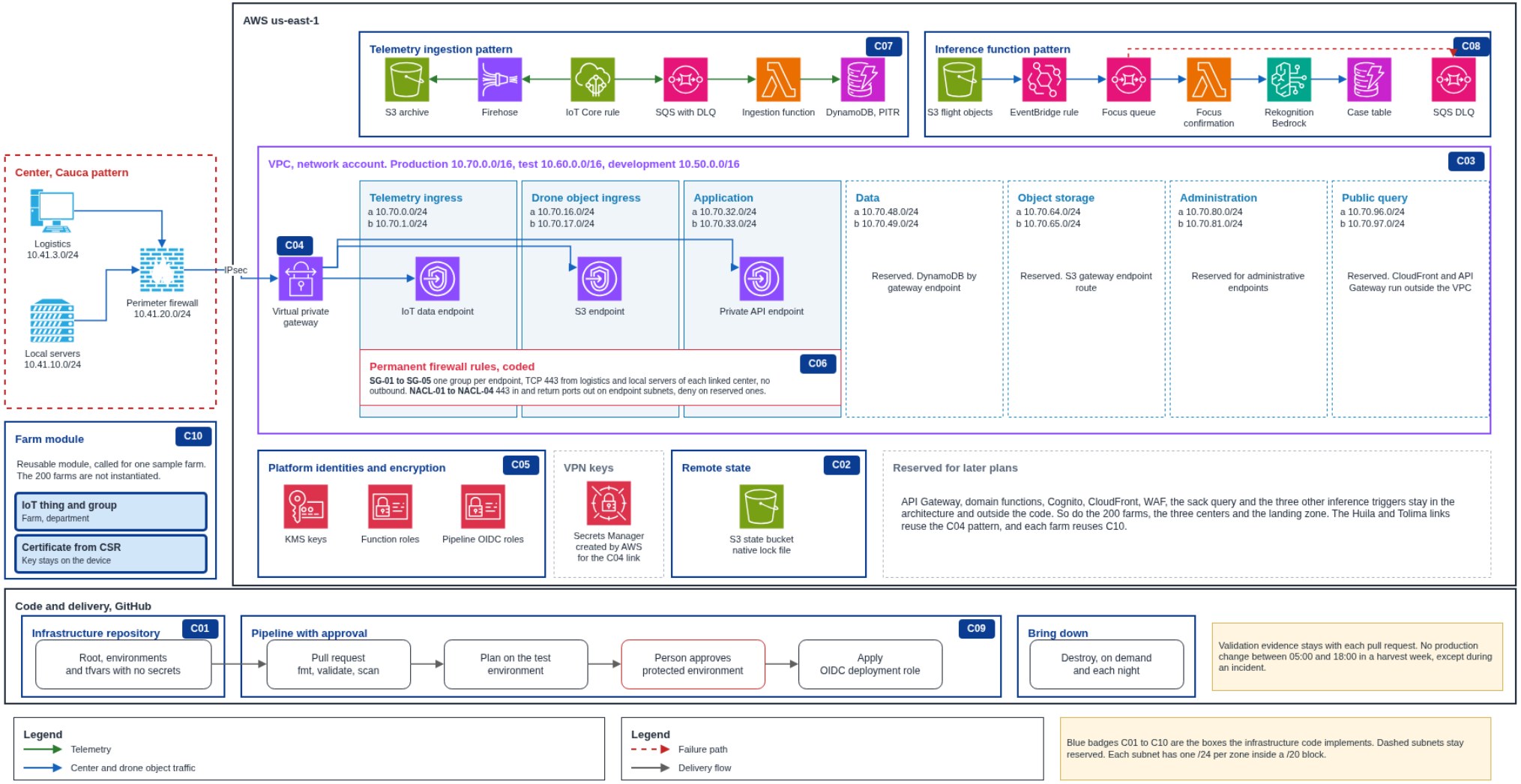
8. Automation and change

The infrastructure code implements the pattern of network, security and one environment within 100 hours. It builds the reusable module of one farm and leaves the 200 farms to that module, as the implementation scope states. The code lives in the infrastructure repository and five versioned module repositories.

Code	Box	Implemented by	Hours
C01	Repository, root and environments	Root configuration and environment variables	12
C02	Remote state	S3 backend with native locking	8

Code	Box	Implemented by	Hours
C03	VPC, subnets and routes	Network module	12
C04	Link to one collection center	Network module	8
C05	Platform identities and encryption	Security module	12
C06	Permanent cloud firewall rules	Security module	8
C07	Telemetry ingestion pattern	Ingestion module	11
C08	Inference function pattern	Function module	12
C09	Pipeline, approval and validation evidence	Pipeline workflows	14
C10	Reusable module of one farm	Farm module	3
	Total		100

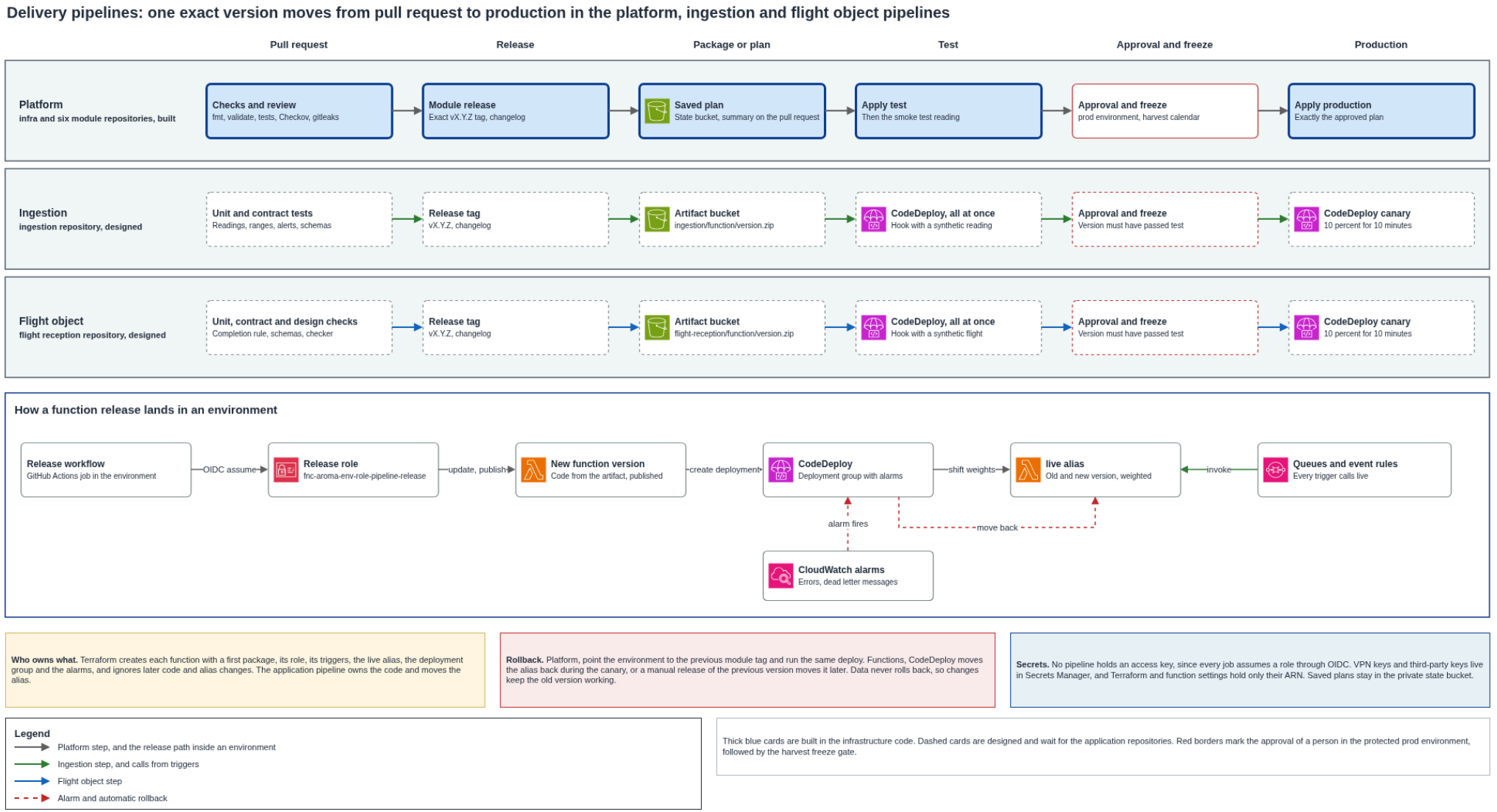
Everything the infrastructure code implements: one environment, boxes C01 to C10, within 100 hours



What the infrastructure code builds, box by box.

The code was planned, applied and destroyed in the test environment through its pipeline, with 127 resources and a smoke test that sent one reading from IoT Core to the readings table. The verification of the infrastructure code holds that evidence.

Development, test and production come from the same code with different variables. A change moves from development to test to production through the pipeline, and production waits for a person's approval, as the environments and promotion design explains. Three delivery pipelines, for the platform, the ingestion code and the flight object code, promote the same version, keep secrets out of the repositories and the state, and go back by redeploying the previous version.



How one version moves from a pull request to production, and how it goes back.

Every production change carries a release note that states what changes, the window, the impact, the approval, the notice to the Federation and the way back. The release notes and change policy hold the two templates and a filled example of each. No production change runs from 05:00 to 18:00 in a harvest week except for an incident, and the production pipeline checks that window before it applies.

9. Workloads

Part A. Farm monitoring

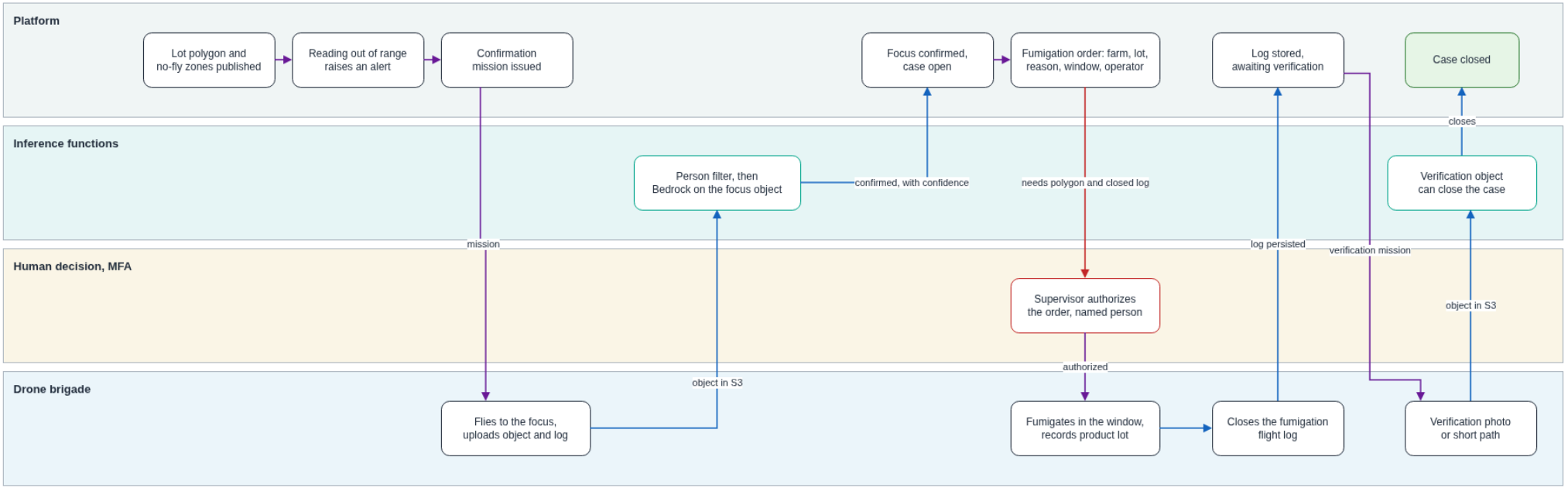
The farm node reads soil and air every 4 hours and the grain every hour while a lot dries, on solar power, installed in one visit, as the sensors and operation and the autonomous node define. The Federation sets the alert ranges as versions in the platform, with no firmware update. Telemetry ingestion stores each reading once, and alert delivery reaches the farmer by push, with SMS only for an urgent drying alert. The farmer application shows Farm, Drying and Alerts, and the Federation reads a monthly pilot report. A visible symptom enters through the drone confirmation of the focus the node marked.

Part B. Drones, pest and fumigation

The drone service combines area work and precision work. Area flights delimit the lot and estimate the harvest, precision flights confirm the focus, fumigate it and verify that same lot, and the confirmed focus moves a lot from one mode to the other, as the operating mode explains. A contracted operator flies, a Federation supervisor in each department authorizes, and the brigades fly a campaign a month in harvest and one every three months outside it, under the roles, permits and campaigns.

Service	What it leaves	What it never does
Delimit the lot	Polygon and no-spray zones around houses, the processing facility, roads and water	No fumigation order leaves without it
Confirm the alert	Photos of the focus and a result with confidence	Authorize a fumigation
Fumigate	Order with farm, lot, reason, window, operator, who authorized, product lot and flight closure	Start without a person's authorization and a closed flight record
Verify	Photos of the treated area and a closable result	Close the order, which only the supervisor does
Estimate the harvest	Estimated sacks for the center, with a range	Create a packing order or spray

From a sensor alert to a closed case: who acts, what each function does, and what stops the flow



Failure and stop conditions

Person in the object. The object moves to a restricted prefix, never reaches inference, and the operator repeats the capture.

Inference fails. The object stays, the state does not change. Three retries, then the dead letter queue and an alarm.

No polygon or no closed log. The order does not go out.

Verification object missing. The fumigation stays open, and the dashboard reports it.

The four event triggers

Trigger	Function persists	Never does
Focus confirmation object **(C08)**	Whether the focus is confirmed	Authorize fumigation
Verification object	Whether the case can close	Change state if the object is missing
Estimation flight object	Estimated sacks for the center	Create a packing order
Telemetry batch	Reading out of threshold	Fix thresholds in code

Each result stores time, confidence and source object in the same case file.

Legend

State and order flow

Object and result flow

Only a person authorizes fumigation. No trigger can do it. C08 marks the function pattern in the first infrastructure plan.

How a crop alert becomes a confirmed focus, an authorized fumigation and a verified, closed case.

The mission is signed and loaded before takeoff, the drone flies with no signal, and the log and objects publish when the link returns, as the drone mission link and the flight objects define. A fumigation never reaches closed while its record is missing.

Part C. Sack and trip

Each 60 kg sack carries a random code and moves through seven states, in preparation, in transit to the center, at the center accepted or rejected, in transit to the port, and delivered accepted or rejected. Every event keeps the date, time and responsible person in an insert-only custody chain, and the state comes from that history, so a late or repeated event does no harm. The public query shows only the buyer view and holds the peak of 30 reads a minute behind CloudFront and its web firewall. The recovery design keeps sack events within 15 minutes of loss and brings the public query back within 4 hours, and a deployment loses no event because senders keep each event until the platform confirms it.

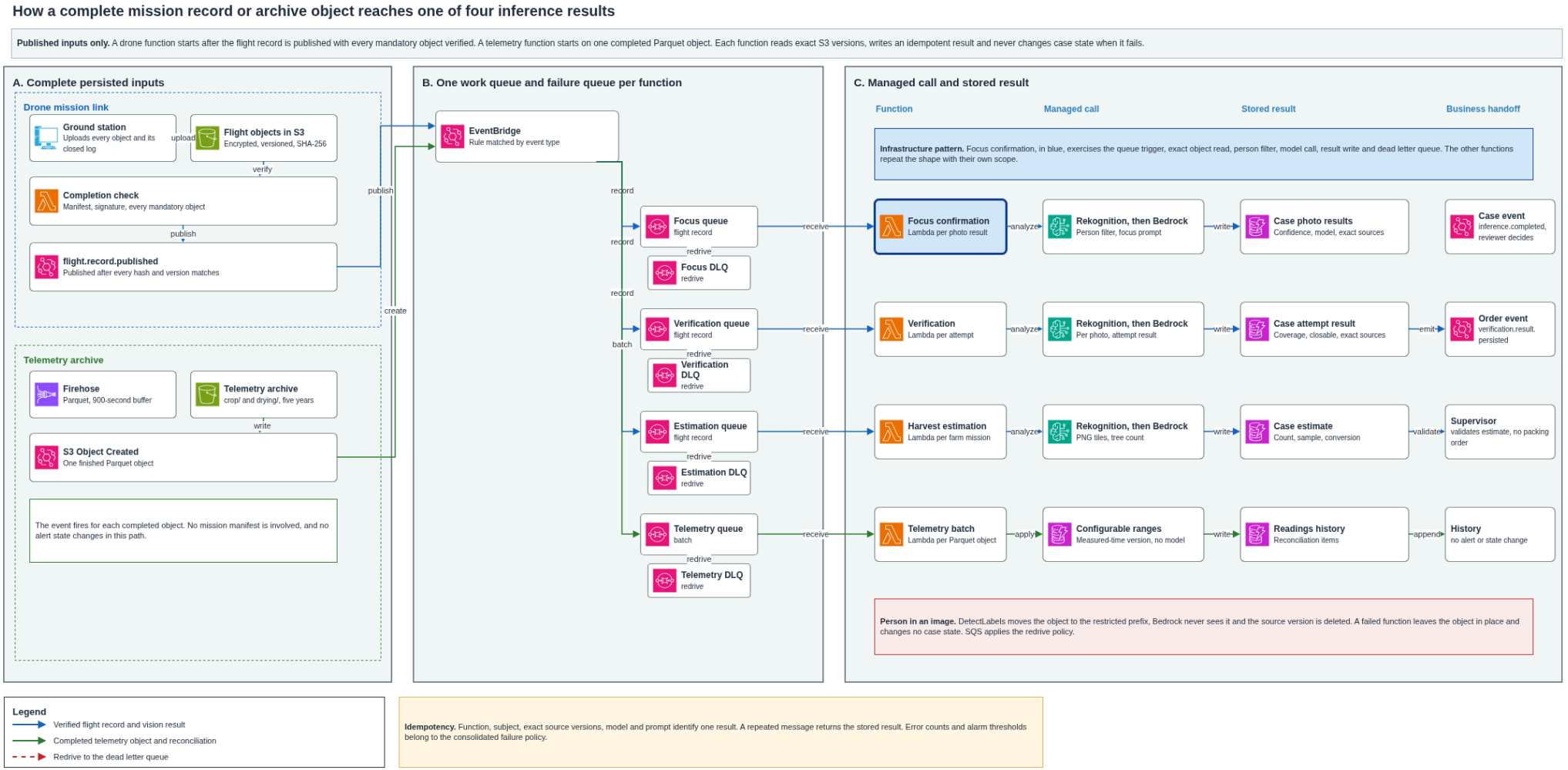
Part D. Collection centers

Logistics stations record reception, inspection, acceptance or rejection and dispatch, and a sack that arrives with no code gets a printed label from the local network, as the collection center operations define for 400 sacks a day in harvest. The WiFi, firewall and LAN of section 7 repeat three times from one pattern.

Part E. Functions and inference

Inference is event driven. A function starts when a complete input is stored, reads that exact version, and writes one result in the case record with its time, confidence and source object, as the inference functions define. The three vision functions call Rekognition to keep people out and Bedrock to analyze the crop, and the telemetry function applies the alert rules. Telemetry lives in DynamoDB and the archive, objects in S3, and sack events and orders in DynamoDB, as persistence and archive describes.

Trigger	Result	Limit
Focus confirmation photos	Focus confirmed or discarded	Never authorizes a fumigation
Verification photos	Whether the order can move toward closure	The state stays if the object never arrives or the call fails
Harvest estimation flight	Estimated sacks for the center	Never creates a packing order
Telemetry batch	Readings out of range under the current rules	The configurable rules stay in force



How a stored flight record or archive object reaches one of the four inference results.

A failed call retries three times, waits in its own dead letter queue with an alarm, and leaves the source object and the case as they were, under the failure policy. Inference runs on the United States profile of the model, and the inference cost is USD 2.76 per 1,000 invocations in a harvest month, USD 1.89 per 1,000 drone images.

10. Observability

The observability design ties a sack, a lot and a flight to their records with shared identifiers, and covers metrics, logs and traces of ingestion, the applications, the database, the three center links and the arrival of flight objects. It keeps six custom metrics, reads the other signals with saved queries, and traces 5 percent of requests.

The operational alarms report nodes and farms silent for more than 24 hours, crop alerts not delivered, flights without a stored record, fumigations that cannot close and inference invocations that failed and left the state unchanged. Each alarm names who answers and the first action.

Objective	Indicator	Target
Crop alert delivery	Live alerts handed off within 5 minutes	99 percent over 30 days
Sack query availability	Lookups answered, with throttles and outages counted as bad	99.5 percent over 30 days

The service levels set both objectives with their error budgets. One operations dashboard and a set of saved cost reports split the pilot by the cost center tag, so farms, drones, collection and tests appear apart.

11. Design system

The pilot has its own identity, with no inheritance from a 2020 manual. The visual identity defines the logo, the isotype and its one-ink and negative versions, a palette with the four sack state badges and the three crop alert levels, the type for phone and web, and the tokens. The design system fixes spacing and grid, the five base components, button, field, alert, sack card and status badge, and rules for outdoor contrast, icons and the buyer view, and applies them on one sheet to the farmer, carrier and buyer views.

12. What happens when something fails

Failure	What the platform does	What is lost
Node loses signal	Stores readings and retries each hour	Nothing in an outage of up to 23 hours. Older readings are dropped and counted
Shared gateway fails	Nodes keep buffering, and an alarm reports the silent vereda	Nothing in an outage of up to 23 hours
Carrier loses data	The application queues positions and custody events, and sends custody first on return	No custody event. Positions older than 24 hours are dropped and counted
Center loses WAN	Staff keep receiving sacks in the application queue, and the printer works on the local network	Nothing, it synchronizes on return

Failure	What the platform does	What is lost
Tunnel fails	The second tunnel of the same connection takes over	Nothing
Function fails	Three retries, then the dead letter queue and an alarm	Nothing. The object stays and the case does not change
Person in a photo	The object moves to a restricted prefix and skips inference	The capture is repeated
Verification never arrives	The fumigation stays open and appears on the dashboard	Nothing
Bad deployment	Pipeline rollback to the previous version	Nothing. Events wait in queues
Table corrupted	Point-in-time restore to a new table, as the recovery design details	At most about 5 minutes, inside the 15-minute objective
Region lost	Rebuild from code and the archive, in a planned effort	Outside the recovery objectives of the pilot

The validation matrix maps each requirement of the challenge to its evidence and follows 18 failure scenarios to the safe state the platform keeps, separating what is built from what is only designed.

13. Scale from 200 to 1,000 farms

Measure	200 farms	1,000 farms	What changes
Device messages a harvest month	108,000	540,000	Nothing, all services scale by themselves
Telemetry cost a harvest month	About USD 1	About USD 5	Linear
Addresses in the cloud plan	Same	Same	Nothing, farms hold no VPC addresses
Ingestion endpoint	One	One	Nothing
Node connections an hour	200	1,000	Nothing, each node keeps its own minute, far below the quota of 3,000 a second

The ingestion sizing document shows the calculation behind these figures and the margin against each service quota.

The stress point at scale sits in the field. A larger pilot needs more shared gateways and more brigades, and neither one changes the cloud.

14. Cost

The pilot cost gathers every cost at current prices, in USD before IVA at the TRM of 3,269.88, and its PDF goes with this document.

Item	Normal month	Harvest month	Following year
Cloud platform, production	181.78	207.07	2,282.52
Links of the three collection centers	229 to 321	229 to 321	2,748 to 3,852
Development and test, with their keys	12.24	12.24	146.88
Field operation of 200 farms	2,176.88	2,176.88	26,122.56
Drone service	3,777.46	20,410.90	111,864.52
Total with the links	**6,377 to 6,469**	**23,036 to 23,128**	**143,164 to 144,268**

The known investment of a farm is USD 467.48, and the proposed monthly amount USD 11.77 with no margin. Inference costs USD 2.76 per 1,000 invocations in a harvest month. The drone service carries almost four fifths of a year, and the cloud platform less than two percent.

The cloud platform of a month breaks down as follows, from the monthly cloud cost.

Item	Normal month, USD	Harvest month, USD
Telemetry ingestion	0.72	0.98
APIs for positions, sacks and queries	1.01	2.31
Inference	1.63	16.26
New objects in S3, for that month	0.11	0.65
Alert delivery, SMS at its upper bound	2.03	10.17
Audit and the small event-driven pieces	0.45	0.45
Operational telemetry, logs, metrics and traces	2.13	2.55
VPN, three connections	109.50	109.50
Interface endpoints, three in two zones	43.80	43.80
KMS, five keys	5.00	5.00
Secrets, four	1.60	1.60
WAF, web ACL and three rules	8.00	8.00
CloudWatch, alarms and the operations dashboard	5.80	5.80
Total	**181.78**	**207.07**

About 84 percent of the normal month is fixed network, the VPN and the endpoints, which keep the center traffic off the internet. The decisions that lower cost are in section 11 of the pilot cost, led by drone missions

directed by the sensors and paid by mission hour.

15. Requirements kept, replaced and retired from 2020

The pilot keeps every capability of 2020 and replaces the connectivity premise that no longer holds. The requirement matrix is reproduced here.

How to read the matrix

Decision	Meaning
Kept	The requirement stands as it was
Replaced	The need stands, and the 2020 way of meeting it does not
Retired	The element leaves the design
Retired as invalid	The value is out of date and cannot be reused, even as a reference

Capabilities

Requirement	Decision	Reason	Affected component
Territory of Cauca, Huila and Tolima	Kept	The pilot keeps its scope	All
Crop and drying monitoring with an alert to the farmer	Kept	It is the core of the service to the farmer	Ingestion, alerts
Harvest and packing record in 60 kg sacks	Kept	Traceability starts here	Sack events
Trip tracking to the center and to Buenaventura	Kept	It proves custody along the route	Carrier application, trip events
Sack lookup by code for farmer, center and buyer	Kept	It is the visible product for the buyer	Sack query
Drone service with its five services	Kept	Each service answers a different need	Drones, objects, inference

Connectivity premises

Requirement or premise	Decision	Reason	Affected component
Data coverage on most farms	Replaced	Some pilot farms lie outside coverage, so a design that needs signal leaves them out	Store-and-forward, shared gateway
Device publishing straight to the ingestion point	Replaced	With no signal the reading would be lost	Node buffer of 24 hours at most
Weekly visits to farms	Retired	Travel is difficult, and a visit must not be a data path	Remote connectivity, drone confirmation

Requirement or premise	Decision	Reason	Affected component
Dependence on a mobile generation being shut down	Retired	The access network changed, and the design must survive the pilot. See the mobile network validity	Node modem on LTE

Sites

Requirement or premise	Decision	Reason	Affected component
Collection centers with no LAN	Replaced	Each center needs a segmented network	LAN, WLAN, firewall
Collection centers with no WAN	Replaced	Reception must reach the platform	100 Mbps link, VPN
Port with no LAN or WAN	Replaced	The Federation deploys nothing in Buenaventura. The shipping company keeps its telemetry, and the platform records only the delivery of the sack	Delivery event

Products and prices

Element	Decision	Reason	Affected component
2020 prices	Retired as invalid	They do not reflect current quotations	Cost document
2020 devices	Retired as invalid	They do not reflect current specifications	Farm node, drone
2020 cloud services	Retired as invalid	Services and prices changed, and the platform uses current ones	Cloud architecture

New requirements with no 2020 equivalent

Requirement	Reason	Component
Local buffer that retains 24 hours at most	Limits how stale a reading can be	Farm node
Mission loaded before takeoff	The drone flies far from any phone	Drone service
Flight log closed before fumigation is authorized	A fumigation needs a verified basis	Drone service
Inference that ends when the result is stored	No compute waits for objects	Functions
Network that grows from 200 to 1,000 farms	The pilot must scale with no redesign	Cloud network, ingestion
Five-year retention, personal data rules and audit	The platform handles personal data	Security, storage

Effect on the design

Three changes shape the whole architecture.

Change	Consequence
Coverage is no longer assumed	Every device buffers, and every flow defines its behavior with no signal
Centers and the port are now part of the design	Connectivity covers five kinds of site
Nothing from 2020 prices or products is reused	Costs and specifications start from current sources

16. Assumptions that shape the design

These assumptions would change the architecture or the cost if they failed. The assumption register lists every assumption of the design.

Assumption	Who confirms	What changes if it fails
4G is available on the routes and at the high points that matter, and an operator enables LTE-M or NB-IoT for the pilot	Operators	The node modem and the data plan change
A shared LoRaWAN gateway reaches the out-of-coverage farms of a vereda from one high point with cellular signal	Radio site study	Those farms need another link, such as satellite, or a data mule for readings as well
Between none and half of the farms are out of cellular coverage, a quarter in the base case	Federation, with the farm list	The investment and the monthly amount move between the cellular and the LoRa farm
One enclosure with wired probes reaches the cultivation and drying zones of a farm	Installer	A second small node per farm, with higher hardware cost
The radio items of the LoRa path, spectrum class, regional plan and dwell time, are confirmed	Regulator and site study	The LoRa path is not deployed and those farms wait for another link
Drone spraying keeps the aerial strips of the regulation, with ground application as the fallback	Legal review	Some lots are treated from the ground and verified by a drone
The drone operator bills by mission hour and uses its staff elsewhere between campaigns	Federation procurement and the operator	A following year rises to about USD 200,738
A harvest lasts four months and the estimation flights spread over them	Departmental committees	Drone and inference lines move in proportion
Each center provider gives a static public address for its link	Link providers	Production cannot be planned until the three addresses exist
The dedicated links cost what the operator reference says	Operator quotes	Links move from USD 64 for shared fiber to 345 a month
The TRM of the price basis and figures before IVA	Federation	A year ranges from 436 to 548 million pesos, and IVA adds up to 19 percent
A Firebase project exists for push notifications	Federation	No push reaches a phone until it exists, and only urgent alerts go out by SMS

Assumption	Who confirms	What changes if it fails
A dedicated short code for SMS in Colombia can be obtained	Federation, through AWS Support	SMS stays on best-effort routes, and the share delivered decides whether the fallback is worth keeping
The harvest weeks of each department are known	Federation	The harvest freeze has no weeks to enforce
Farmers accept phone sign-in	Federation	A simpler sign-in flow is designed
The first restore of the sack tables fits the budget of 3 hours 30 minutes	First restore exercise	The procedure changes to meet the 4-hour objective

17. Diagrams

The complete diagram is page 00 of the diagram file. It shows the addressing in zones C and D3, the perimeter firewall of each center and the permanent cloud firewall rules, the store-and-forward buffer of each farm in zone B, the inference functions in zone D2, and the boxes C01 to C10 that the code builds. Page 04 carries the flow from pest to fumigation. The other pages detail one subject each.

Page	Question it answers	Section
00 Platform overview	What is the whole platform	3
01 End to end	Which path, boundary and identity does each flow have	4
02 Implementation scope	What does the infrastructure code build, box by box	8
03 Connectivity and centers	How does each site connect, and what happens with no signal	7
04 Pest to fumigation	How does an alert become a closed case	9
05 Landing zone	Which accounts exist, and what applies from the root	5
06 Environments	How does a change reach production, and how do environments go up and down	8
07 Farm connection sequence	How does a farm node connect, lose its link and recover	7
08 Sack state model	Which states can a sack pass through, and what does a rejection do	9
09 Center WLAN	Which SSID lands on which VLAN, and what can each zone reach	7
10 Drone mission link	How is a drone mission preloaded, flown with no signal and published	9
11 Audit evidence path	How does evidence of a sensitive event reach a store that no account can alter	5
12 Lot geometry lifecycle	Which states can a lot polygon pass through, and which strips keep spraying away	9
13 Center operation	How does a center receive, label and dispatch sacks	9
14 Crop case states	How does a cultivation alert become a focus confirmed or discarded by a person	9
15 Alert delivery sequence	How does an alert reach the farmer, and how is each step recorded	9

Page	Question it answers	Section
16 Fumigation order states	Which states can a fumigation order pass through, and which moves only a person can make	9
17 Public query	How does a buyer read one sack, and what limits bulk reads	9
18 Center cloud link	How does a center reach the cloud, and what happens when the link fails	7
19 Inference functions	How does a stored record reach one of four inference results	9
20 Signals and correlation	Which identifiers travel with a sack, a lot and a flight, and where is each signal read	10
21 Flight object transport	Where does each flight object live, and which events carry it to a published record	9
22 Delivery pipelines	How does one version move from a pull request to production, and how does it go back	8

18. Documents by subject

Subject	Documents
Governance	Landing zone, cloud selection, requirement matrix, tags, policies and budgets, audit
Security	Identity and access, encryption, secrets and revocation, personal data and retention, firewall policy, inference data egress
Network and connectivity	Farm connectivity, mobile network validity, LoRa out of coverage, drone mission link, carrier tracking, port delivery, center LAN, center WLAN, center WAN sizing, center cloud link, cloud addressing
Automation and change	Implementation scope, environments and promotion, delivery pipelines, release notes, infrastructure repository
Farm monitoring	Sensors and operation, autonomous node, alert ranges, telemetry ingestion, ingestion sizing, alert delivery, farmer history, alert to drone confirmation
Drones	Operating mode, roles, permits and campaigns, lot boundaries, focus confirmation, fumigation order, verification, harvest estimation, flight objects
Sack and centers	Sack identity, sack states, custody chain, views by actor, public query, carrier application contract, center operations
Functions and data	Inference functions, failure policy, persistence and archive, recovery
Observability	Observability, operational alarms, service levels, dashboards
Cost	Pilot cost, price basis, cloud cost, farm hardware and monthly amount, drone campaign cost, inference cost
Design system	Visual identity, design system
Validation, decisions and assumptions	Validation matrix, decision records, assumption register, planning workbook